
FLAVOURBENCH: An Executable Benchmark for Culinary Reasoning Without a Model Judge

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Abstract

1 FLAVOURBENCH is an executable benchmark for culinary reasoning across frontier
2 language models, scored against answer keys computed by EPICURE, a versioned
3 culinary system. Four deterministic, read-only operations compile a common 32-
4 task panel before any evaluated model is called; no human or model judge scores the
5 test. We run 20 current endpoints on every task. Model-only exact-choice accuracy
6 is the FlavourBench Score; missing outputs count as incorrect and completion is
7 reported separately. GPT-5.6 Sol Pro leads this fixed panel with 20/32 correct, or
8 62.5% (Wilson 95% interval 45.3–77.1%). The leading intervals overlap, and exact
9 paired tests against both 19/32 runners-up give $p = 1.000$, so the release supports
10 an exact score snapshot rather than a resolved fine-grained ordering. A secondary
11 named-operation condition measures execution with EPICURE available. It reaches
12 100% for 14 of 19 responding endpoints and is therefore an integration diagnostic,
13 not a second leaderboard. A content-addressed release binds every task, answer key,
14 route, response, completion state, and Epicure trace; its dependency-free verifier
15 reconstructs the score table and figures offline.

16 1 Introduction

17 Benchmarks are easiest to trust when their ground truth is external to the models being tested. Code
18 can be executed, database states can be inspected, and games can be scored. Culinary reasoning
19 rarely has that luxury. Ingredient substitutions, pairings, technique choices, and cuisine associations
20 are commonly assessed by another language model or by a small set of human preferences. The
21 evaluator then becomes part of the construct.

22 FLAVOURBENCH starts from EPICURE, a read-only culinary runtime with 1,790 ingredients repre-
23 sented in a 300-dimensional space [1, 2]. Four runtime operations are converted into four-choice
24 questions. The compiler records each answer before it contacts an evaluated model.

25 The design asks one ranking question and one diagnostic question. The FlavourBench Score measures
26 how often a model endpoint returns EPICURE’s answer without runtime access. The same endpoint
27 then receives the task with one read-only EPICURE call, providing a controlled execution check.
28 Every model receives every task in both conditions. Only the model-only score determines rank.

29 Our release evaluates 20 current endpoints, including GPT-5.6 Sol, Terra, and Luna; Claude Fable,
30 Opus, and Sonnet 5; Gemini 3.1 Pro and 3.6 Flash; Grok 4.5; Kimi K3; Qwen 3.8 Max; GLM 5.2;
31 two DeepSeek V4 variants; MiniMax M3; Nemotron 3 Ultra; Mistral Large 2512; Tencent HY 3;
32 and Cohere Command A Plus and Command A Reasoning. Kimi and both Cohere models use direct
33 provider routes. Every other endpoint uses an exact, no-fallback OpenRouter route selected before
34 the run.

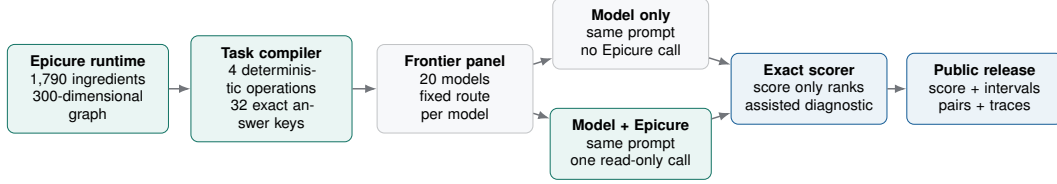


Figure 1: The FLAVOURBENCH pipeline. EPICURE fixes each answer before evaluation. The Model only arm produces the ranked FlavourBench Score. The matched Model + Epicure arm tests execution with the answer-generating system available and does not affect rank. Both reconstruct from the same public release.

Family	Epicure operation	Exact target	Example query	Tasks
Substitution	<code>neighbors</code>	first ranked neighbour	top neighbour for tomato	8
Composition	<code>pairing_score</code>	four-decimal affinity	tomato + basil	8
Cookability	<code>compare_on_axis</code>	larger flavour projection	coffee vs. chocolate on bitter	8
Evidence	<code>cultural_profile</code>	highest cuisine cosine	cuisine direction for kimchi	8

Table 1: The task set is balanced by construction. Each family maps one deterministic EPICURE operation to a four-choice answer. Chance accuracy is 25%.

35 The paper makes three contributions:

- 36 1. an automated culinary benchmark whose questions and answer keys come from a versioned
- 37 executable system rather than a model judge;
- 38 2. a common-task frontier evaluation with 640 matched pairs, exact missing-response penalties, and
- 39 a public 20-row leaderboard;
- 40 3. a matched, named-operation diagnostic that exposes tool-integration failures and task-family
- 41 interactions without changing the ranking score.

42 EPICURE has two roles. It fixes the answer key, then becomes the intervention in the second arm.

43 This keeps the leaderboard and the Epicure comparison connected without combining them into an

44 arbitrary composite score.

45 2 Epicure reference answers

46 2.1 A versioned reference

47 For FLAVOURBENCH, ground truth is the output of a specific version of EPICURE. It is not a claim

48 about universal culinary preference. The release binds the application, evidence bundle, release

49 identifier, ingredient count, and embedding dimensionality by hash. A different runtime produces a

50 different task-set hash.

51 This definition makes every answer independently recomputable. It also gives the score a direct

52 interpretation: accuracy is agreement with the published culinary runtime under the published task

53 construction. No second model decides which response sounds better.

54 2.2 Four balanced task families

55 The compiler creates eight tasks in each of four families. Seeds and ingredient pairs are fixed before

56 evaluation. For every task, EPICURE is called once to obtain the reference result. The correct answer

57 is extracted programmatically, three distractors are constructed without model generation, and the

58 four options are shuffled with a task-ID-derived seed.

59 The task families deliberately mix retrieval and computation. Neighbour and cuisine questions test

60 whether a model knows relationships encoded by the graph. Pairing questions require an exact

61 numeric result that is difficult to guess. Axis questions test a comparative claim rather than a

62 memorised scalar. Together they prevent the leaderboard from reducing to one operation or one kind

63 of culinary trivia.

64 2.3 Prompt and answer contract

65 Each prompt names the relevant operation and gives four choices. Models may explain briefly, but
66 the last answer line must be FINAL_CHOICE: X. The parser accepts A, B, C, or D on that marker
67 line. A missing marker, abnormal provider finish, timeout, or absent response scores zero. The rule
68 fits in the standalone verifier shipped with the release.

69 The complete task set contains no hidden human label and no model-authored item. The benchmark is
70 therefore suitable for automatic reruns, CI-style regression tests, and training loops where thousands
71 of candidate responses must be scored consistently.

72 3 Evaluation design

73 3.1 One score and one matched intervention

74 For model m and task t , let M_{mt} equal one when the Model only response finishes normally and
75 matches the EPICURE answer. Let E_{mt} be the same indicator for Model + Epicure. The prompt,
76 route, temperature, top- p , seed request, output limit, and answer parser are identical. Only access to
77 EPICURE changes.

78 Each Model + Epicure response may make one read-only call to the operation named in the prompt.
79 The runner records the operation, arguments, latency, error flag, and a hash of the returned payload.
80 The returned data is bounded before it enters model context. Automatic provider fallback is disabled,
81 so a route failure remains a failure for that route.

82 The single ranking metric is the FlavourBench Score:

$$F_m = 100 \frac{1}{32} \sum_t M_{mt}. \quad (1)$$

83 Every family contains eight tasks, so ordinary accuracy already gives each family equal weight.
84 Missing or invalid responses remain in the denominator as incorrect. The paper and public Space give
85 equal values of F_m the same score rank; model name orders rows within a tie. No assisted metric
86 breaks a FlavourBench Score tie.

87 Epicure Gain is the matched difference

$$G_m = 100 \frac{1}{32} \sum_t (E_{mt} - M_{mt}), \quad (2)$$

88 reported in percentage points. G_m is not part of the FlavourBench Score. The release also keeps the
89 four paired outcomes for every task: both correct, Model only correct, Model + Epicure correct, or
90 neither correct.

91 3.2 Completion and uncertainty

92 Completion is the fraction of assigned responses that end normally with a parseable final marker.
93 The score still counts missing responses as incorrect. Reporting completion beside the score keeps
94 endpoint failure separate from a wrong culinary answer. A zero score with zero completions is not
95 evidence of zero model capability. For each 32-task accuracy we report a 95% Wilson interval as
96 a descriptive uncertainty indicator. The panel is deliberately balanced rather than randomly drawn
97 from all culinary questions, so the interval is not a population guarantee.

98 The benchmark is designed for visible differences, not hundredths of a point. One answer changes
99 the score by 3.125 points. Exact paired McNemar tests comparing the 20/32 leader with each 19/32
100 runner-up give $p = 1.000$ and $p = 1.000$. These unadjusted descriptive tests, the overlapping
101 intervals, family breakdowns, and task matrix all point to the same reading: this release resolves
102 broad separation, not the order of adjacent leaders.

103 3.3 Twenty-model frontier panel

104 The panel was fixed before the final calls. It includes three GPT-5.6 tiers rather than treating one
105 deployment as the entire OpenAI family; three Claude 5 tiers; two Gemini tiers; two DeepSeek

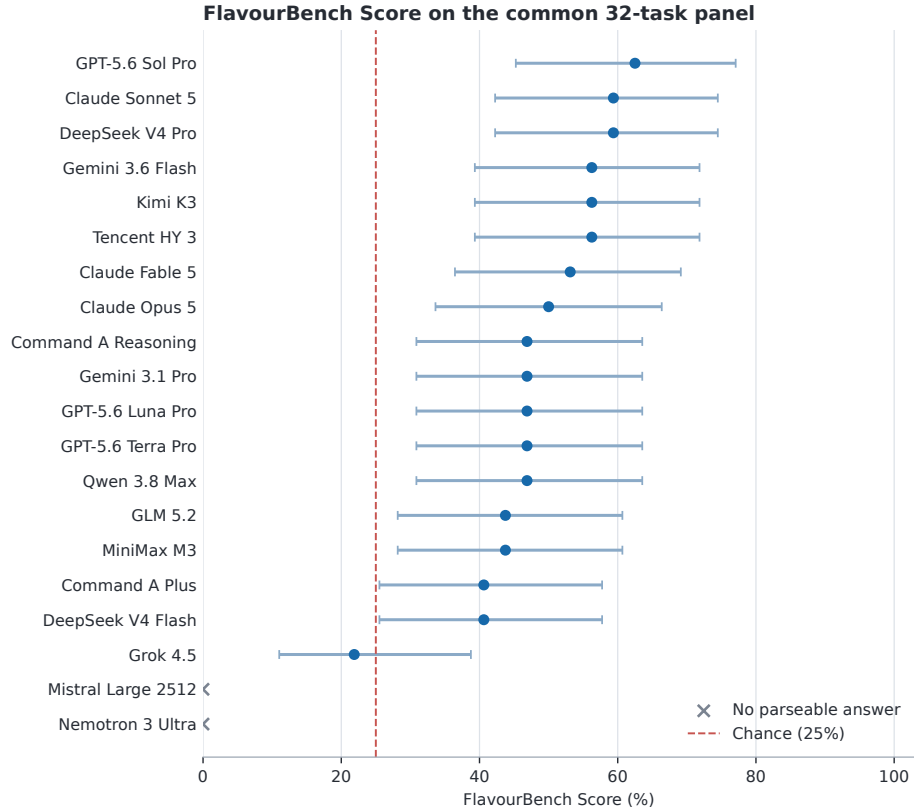


Figure 2: FlavourBench Score against EPICURE answer keys. Points are exact accuracy over the common 32-task panel; bars are descriptive 95% Wilson intervals. Crosses mark endpoints with no parseable model-only answer. The dashed line is four-choice chance.

V4 routes; both current Cohere Command A routes; and a broad set of current independent and open-weight families. Kimi K3 runs through Kimi’s Anthropic-compatible endpoint. Command A Plus and Command A Reasoning run through Cohere V2 Chat. Qwen 3.8 Max and GLM 5.2 use explicit OpenRouter routes in the automated batch [5–7, 14, 18, 19]. The exact 20-row route table appears in Appendix C.

4 Results

4.1 The FlavourBench Score leaderboard

GPT-5.6 Sol Pro leads the fixed panel with 20/32 correct and a FlavourBench Score of 62.5 (Wilson 95% interval 45.3–77.1). Table 2 gives the score-only ranking, correct counts, intervals, and parsed-answer counts. The top three differ by one answer, their intervals overlap substantially, and the paired tests above do not separate the leader from either runner-up. Across all 19 leader comparisons, Holm adjustment separates only 3 rows, Grok 4.5, Mistral Large 2512, Nemotron 3 Ultra; two are DNF endpoints. Models with no parseable model-only answer are marked DNF rather than interpreted as the least capable systems.

The answer key exists before the model call, every endpoint receives the same questions, and no judge is sampled after the fact. Missing outputs count as incorrect in the operational score but remain visible in the “Parsed” column. Mistral Large 2512 returned 32 normal model-only responses, but none contained a parseable final choice. Nemotron 3 Ultra returned no response in either condition. Both zeroes are operational outcomes rather than capability estimates.

Score rank	Model	Correct	FB Score	Wilson 95%	Parsed
1	GPT-5.6 Sol Pro	20/32	62.5	45.3 to 77.1	31/32
2	Claude Sonnet 5	19/32	59.4	42.3 to 74.5	32/32
2	DeepSeek V4 Pro	19/32	59.4	42.3 to 74.5	32/32
4	Gemini 3.6 Flash	18/32	56.2	39.3 to 71.8	32/32
4	Kimi K3	18/32	56.2	39.3 to 71.8	32/32
4	Tencent HY 3	18/32	56.2	39.3 to 71.8	32/32
7	Claude Fable 5	17/32	53.1	36.4 to 69.1	30/32
8	Claude Opus 5	16/32	50.0	33.6 to 66.4	32/32
9	Command A Reasoning	15/32	46.9	30.9 to 63.6	30/32
9	Gemini 3.1 Pro	15/32	46.9	30.9 to 63.6	30/32
9	GPT-5.6 Luna Pro	15/32	46.9	30.9 to 63.6	32/32
9	GPT-5.6 Terra Pro	15/32	46.9	30.9 to 63.6	32/32
9	Qwen 3.8 Max	15/32	46.9	30.9 to 63.6	28/32
14	GLM 5.2	14/32	43.8	28.2 to 60.7	32/32
14	MiniMax M3	14/32	43.8	28.2 to 60.7	31/32
16	Command A Plus	13/32	40.6	25.5 to 57.7	32/32
16	DeepSeek V4 Flash	13/32	40.6	25.5 to 57.7	24/32
18	Grok 4.5	7/32	21.9	11.0 to 38.8	10/32
DNF	Mistral Large 2512	0/32	0.0	0.0 to 10.7	0/32
DNF	Nemotron 3 Ultra	0/32	0.0	0.0 to 10.7	0/32

Table 2: The score-only public leaderboard. “Correct” and “FB Score” use model-only answers; equal scores share a score rank. Wilson 95% is a descriptive interval for the fixed 32-task panel, and “Parsed” reports responses with a valid final choice. DNF means that the endpoint returned no parseable model-only answer.

Statistical indicator	Observed value	Reading
Score resolution	1/32 = 3.125 points	Adjacent rows often differ by one answer.
Leader uncertainty	62.5%; Wilson 95% 45.3–77.1%	Exact for this panel; broad descriptive interval beyond it.
Leader vs. 19/32 runners-up	Exact paired $p = 1.000$ and $p = 1.000$	The fixed panel does not separate the top three.
Leader vs. full panel	3/19 comparisons significant after Holm adjustment	Only broad operational separation is resolved.
Assisted ceiling	14/19 endpoints at 100%	Useful for integration failures, not rank discrimination.
Score–gain correlation	$r = -0.990$	Near-perfect inverse dependence; gain is largely the distance to the assisted ceiling.

Table 3: The statistical reading of this release. Intervals and tests are descriptive because the 32 tasks form a fixed balanced panel rather than a random sample of all culinary questions.

125 4.2 What Epicure changes

126 The assisted condition is deliberately secondary. Among the 19 endpoints that returned assisted
127 completions, 14 score 100% and mean accuracy is 98.8%. FlavourBench Score and Epicure Gain
128 correlate $r = -0.990$ because a near-100% assisted ceiling makes gain approximately $100 - F_m$.
129 Gain therefore adds little ranking information and does not appear in the primary table.

130 The assisted prompt names the relevant operation, whose output is also used to compile the answer
131 key. Near-ceiling assisted accuracy is therefore expected. The condition tests whether an endpoint can
132 call the specified operation, read its output, and preserve the answer contract. It is not an independent
133 validation of EPICURE, a second capability score, or a test of open-ended tool discovery.

134 4.3 Families explain aggregate ties

135 The four task families have different demands. Pairing questions require an exact four-decimal value;
136 substitution and cultural questions can sometimes be answered from broad food knowledge; axis
137 comparisons require a directional decision in EPICURE’s latent space. Figure 4 shows that similar
138 total scores can arise from different profiles.

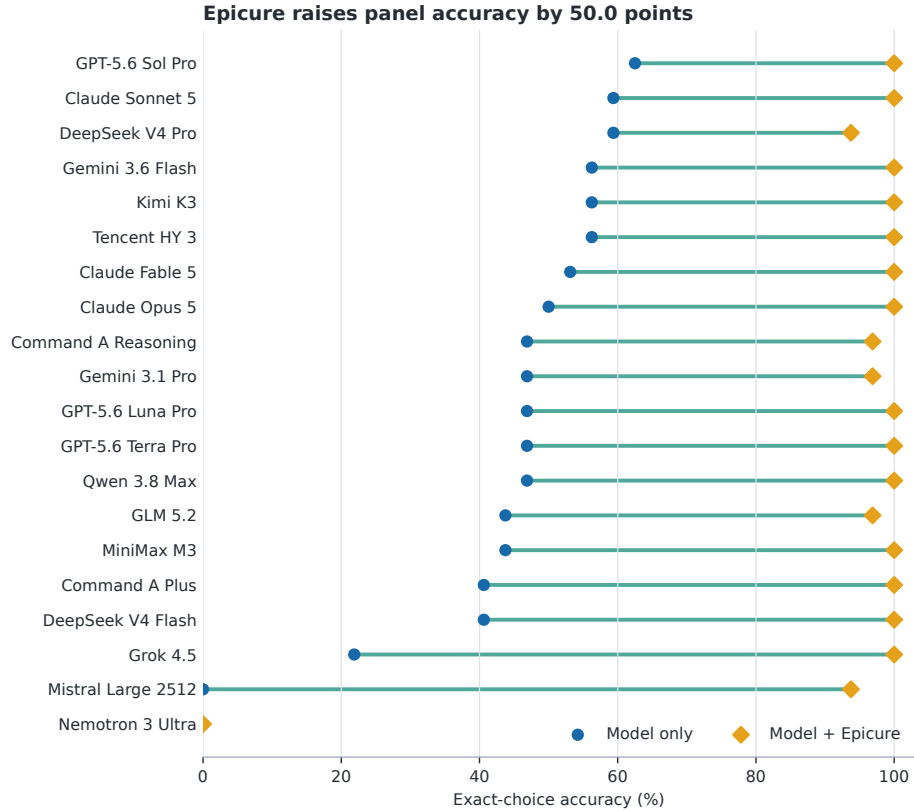


Figure 3: Model-only and named-operation assisted accuracy over the same 32 tasks. This secondary diagnostic shows the assisted ceiling and endpoint failures; it does not determine rank.

Family	Epicure operation	Tasks	Model only	Model + Epicure
Substitution	<code>neighbors</code>	8	18.1	94.4
Composition	<code>pairing_score</code>	8	40.6	94.4
Cookability	<code>compare_on_axis</code>	8	62.5	92.5
Evidence	<code>cultural_profile</code>	8	54.4	94.4

Table 4: Panel-average performance by family and the corresponding reference operation.

139 4.4 Every model and task pair is inspectable

140 Aggregate bars can hide a benchmark bug or a single pathological task. Figure 5 therefore publishes
 141 all 640 paired outcomes. Blue cells are stable knowledge: both conditions are correct. Gold cells
 142 are Epicure wins: Model only is wrong and Model + Epicure is correct. Red cells show the reverse
 143 outcome. Grey cells show questions neither condition solved.

144 4.5 Cohere, Kimi, Qwen, and route fidelity

145 Both Cohere models are full members of the public panel. Command A Plus has score rank 16
 146 with a FlavourBench Score of 40.6; Command A Reasoning has score rank 9 at 46.9. Kimi K3 has
 147 score rank 4 through the direct Kimi endpoint. Qwen 3.8 Max has score rank 9 through an exact
 148 OpenRouter route, and GLM 5.2 has score rank 14. No route changes during generation.

149 The two direct Cohere routes each produced all 64 assigned arms. Their returned identifiers are `comm`
 150 `and-a-plus-05-2026` and `command-a-reasoning-08-2025`, respectively; the direct Kimi route

likewise produced all 64 arms under returned identifier k3. The offline verifier checks these provider, identity, and arm-count contracts before accepting the release.

This matters because a model name is not a transport contract. The release stores the requested model, returned model, provider, route, generation ID, latency, token counts, and response hash. Direct and routed endpoints can therefore be analysed separately without pretending they are interchangeable deployments.

4.6 Score and response time

Figure 6 plots FlavourBench Score against median response time. Marker colour identifies the execution route. Accuracy and latency are separate operational dimensions: faster endpoints do not necessarily score higher, and direct-provider routes are not pooled with routed deployments.

5 Using the score as a training signal

The same environment exposes four binary indicators for a generated trajectory τ : r_{answer} for a correct answer, r_{format} for a parseable final marker, r_{call} for a successful Epicure call, and r_{align} for a call matching the reference operation. The public leaderboard uses r_{answer} in Model only. A future training objective could use

$$r(\tau) = r_{\text{answer}}r_{\text{format}} + \lambda r_{\text{answer}}r_{\text{call}} + \mu r_{\text{align}}, \quad (3)$$

where each component is a binary indicator and λ, μ set the value of successful and reference-matching calls. Requiring a correct answer prevents irrelevant calls from earning the main reward. The interface could support rejection sampling, preference optimisation, or reinforcement learning on new ingredient seeds. This paper does not train a checkpoint; such a study should keep the public leaderboard as a final evaluation split.

6 Reproducibility and release

The public release is one JSON document with hash 197b84d75cca... It contains the 20 route records, 32 full prompts, answer keys and reference EPICURE results, 1,280 observation rows, final answer markers, response times, and bounded Epicure-trace projections. 1195 provider responses were observed; 85 missing arms remain explicit zero-scored rows.

The leaderboard can be reconstructed without a network connection, provider key, GPU, or running EPICURE service:

```
python3 -I reproduce_epicure_native.py \
  --release generated/epicure-native/\
  epicure-native-release.json
```

The verifier rejects duplicate JSON keys, non-finite numbers, a changed content address, an incomplete $20 \times 32 \times 2$ grid, or any leaderboard value that differs from recomputation. The paper tables and figures are generated from the same release, so a changed score cannot leave a stale chart behind.

Reproducing the original hosted calls is a different operation from replaying the result. The release records exact routes and request policies, but provider deployments can change. A fresh frontier run should generate a new manifest and release rather than overwrite this snapshot.

7 Discussion

What the score means. The FlavourBench Score answers a narrow operational question: how often did this deployed endpoint return the answer fixed by the published EPICURE runtime? It includes endpoint availability because missing responses score zero. The completion columns reveal that component. The score is not a universal measure of cooking skill, but it is exact, reproducible, and independent of a model judge.

Benchmark	Scope	Primary target	Paired tool?
Chatbot Arena	general chat	pairwise human preference	No
WildBench	general chat	task-specific checklist	No
BFCL	function calling	executable call match	No
τ -bench	interactive agents	exact goal state	No
DiningBench	multimodal food	dish and nutrition labels	No
FAM-Bench	food-as-medicine	expert suitability label	No
FLAVOURBENCH	culinary reasoning	versioned executable EPICURE operation	Yes

Table 5: FLAVOURBENCH complements broad, tool-use, and food-domain benchmarks. Its distinctive unit is the same endpoint and task measured as Model only and Model + Epicure; “paired tool” denotes that matched intervention rather than ordinary tool availability.

Do we need humans? Not for this leaderboard. Humans are useful when the target is preference, creativity, safety, or the quality of an open-ended recipe. They are unnecessary when the question is whether a model’s choice matches a deterministic runtime result. FLAVOURBENCH uses that distinction to automate the core ranking while leaving subjective culinary evaluation as a separate track.

Why run Model + Epicure? The condition is a named-operation execution test. Because EPICURE both generated the answer key and supplies the relevant runtime output, its near-ceiling accuracy is expected. The condition still catches tool-call, argument, parsing, and answer-contract failures, but it neither ranks models nor independently validates EPICURE. The dumbbell and pair matrix expose those failures task by task.

Scaling the benchmark. The task compiler can expand the seed lists while preserving the four operations. The next release should reserve unseen ingredients and pairs, remove near-duplicate questions, and measure stability across runtime versions. As a simple scale guide near 50% accuracy, 100 independent tasks give about a 10-point 95% half-width and 400 give about five points; paired rank claims should be powered on the observed task covariance. The current panel is a frontier snapshot and a reproducible template for that larger hidden set.

8 Related work

HELM argues for explicit scenarios and metrics, while recent benchmark audits emphasise validity, contamination, and transparent reporting [15, 23]. Chatbot Arena and WildBench evaluate broad answer quality through preferences or checklists [3, 16]. FLAVOURBENCH instead targets an executable domain model and needs no judge for its primary score.

ToolLLM, BFCL, and τ -bench make tool invocation or environment state observable [20, 21, 28]. FLAVOURBENCH adds a matched Model only condition, so EPICURE becomes an intervention rather than a requirement shared by every response. LiveBench, LiveCodeBench, GPQA, MMLU-Pro, and SWE-bench illustrate complementary strategies for reducing contamination or grounding answers in difficult tasks and executable outcomes [11, 12, 22, 25, 26].

Culinary datasets such as FlavorDB and Recipe1M organise ingredient and recipe evidence [9, 24]. Recent food benchmarks emphasise other constructs: NutriBench estimates carbohydrates from meal descriptions; FoodBench-QA grounds nutrition tasks in databases and ontologies; DiningBench tests multimodal dish perception and nutrition reasoning; and FAM-Bench evaluates condition-aware food-as-medicine decisions [8, 10, 13, 17]. CookingSense and planning-oriented culinary work test commonsense and procedures [4, 27]. FLAVOURBENCH targets a complementary construct: ingredient-space flavour decisions whose targets are compiled from a callable, versioned runtime, with access to that runtime measured as a matched intervention.

227 9 Limitations

228 The 32-task panel resolves broad differences, not adjacent leaders: one answer moves the score by
229 3.125 points and the leading intervals overlap. The English ingredient seeds cover four EPICURE
230 operations, hosted routes are a dated snapshot, and public prompts may eventually be memorised.
231 Agreement with EPICURE inherits the runtime’s representation choices; it is not a food-safety or
232 cultural-authenticity certification. A larger hidden panel, additional operations, and repeated route
233 snapshots are the immediate next tests.

234 10 Conclusion

235 FLAVOURBENCH turns a versioned culinary runtime into an executable benchmark for frontier
236 language models. GPT-5.6 Sol Pro leads this 20-model, 32-task snapshot at 62.5%, while the
237 overlapping leading intervals make the close group explicit. Scores, completions, prompts, responses,
238 and traces all reconstruct from the released 1,280-arm grid. The result is a simple automated
239 leaderboard with a separate named-operation diagnostic, not a model judge hidden behind another
240 score.

241 A Real prompts, answers, and tool calls

242 A benchmark paper should show what was actually asked. The four cases below are selected
243 mechanically, one per family. Where possible, the builder chooses a model and task that is wrong in
244 Model only and correct in Model + Epicure. Each case includes the full prompt, both final answers,
245 the actual operation and arguments, and the exact answer key.

246 **Case 1: Substitution with GPT-5.6 Sol Pro. Prompt.** FlavourBench Epicure-native exact task.
247 Use the read-only culinary evidence tools if they are available; otherwise answer from your own
248 knowledge.

249 According to Epicure’s ‘neighbors’ result for ‘tomato’ with top_k=4, which ingredient is ranked first?

250 Choices: A. oregano B. bell_pepper C. olive_oil D. onion

251 In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL_CHOICE:
252 X’, replacing X with A, B, C, or D.

253 **Model only.** FINAL_CHOICE: D **Model + Epicure.** FINAL_CHOICE: B

254 **Epicure call.** neighbors({"ingredient": "tomato", "top_k": 4}). The answer key is
255 FINAL_CHOICE: B.

256 **Epicure result.** {"ingredient": "tomato", "neighbors": [{"name": "bell_pepper", "rank":
257 1, "sim": 0.4799}, {"name": "olive_oil", "rank": 2, "sim": 0.4786}, {"name": "oregano",
258 "rank": 3, "sim": 0.4646}, {"name": "onion", "rank": 4, "sim": 0.4567}]}

259 **Case 2: Composition with DeepSeek V4 Pro. Prompt.** FlavourBench Epicure-native exact task.
260 Use the read-only culinary evidence tools if they are available; otherwise answer from your own
261 knowledge.

262 What exact four-decimal ‘pairing_score’ does Epicure return for ‘tomato’ and ‘basil’?

263 Choices: A. 0.2707 B. 0.2407 C. 0.3207 D. 0.1907

264 In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL_CHOICE:
265 X’, replacing X with A, B, C, or D.

266 **Model only.** FINAL_CHOICE: B **Model + Epicure.** FINAL_CHOICE: A

267 **Epicure call.** pairing_score({"ingredient_a": "tomato", "ingredient_b":
268 "basil"}). The answer key is FINAL_CHOICE: A.

269 **Epicure result.** {"all_pairs_range": {"median": 0.092, "p10": 0.0197, "p25": 0.0521, "p75":
270 0.1392, "p90": 0.1899}, "pairing_score": 0.2707, "percentile_label": "very high (>=p90)",
271 "resolved_a": "tomato", "resolved_b": "basil"}

272 **Case 3: Cookability with GPT-5.6 Sol Pro. Prompt.** FlavourBench Epicure-native exact task.
273 Use the read-only culinary evidence tools if they are available; otherwise answer from your own
274 knowledge.

275 Which statement matches Epicure’s ‘compare_on_axis’ result for ‘coffee’ versus ‘chocolate’ on
276 ‘cf_bitter’?

277 Choices: A. coffee has the higher projection B. the projections are equal to four decimals C. Epicure
278 cannot resolve one or both ingredients D. chocolate has the higher projection

279 In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL_CHOICE:
280 X’, replacing X with A, B, C, or D.

281 **Model only.** FINAL_CHOICE: A **Model + Epicure.** FINAL_CHOICE: D

282 **Epicure call.** compare_on_axis({"axis": "cf_bitter", "ingredient_a": "coffee",
283 "ingredient_b": "chocolate"}). The answer key is FINAL_CHOICE: D.

284 **Epicure result.** {"axis": "cf_bitter", "axis_range": {"p10": -0.0877, "p90": 0.1429},
285 "delta": 0.0835, "label_a": "moderate", "label_b": "high", "projection_a": 0.0076,
286 "projection_b": 0.0911, "resolved_a": "coffee", "resolved_b": "chocolate"}

287 **Case 4: Evidence with Grok 4.5. Prompt.** FlavourBench Epicure-native exact task. Use the
 288 read-only culinary evidence tools if they are available; otherwise answer from your own knowledge.
 289 Which cuisine direction has the highest cosine score in Epicure’s ‘cultural_profile’ result for ‘miso’?
 290 Choices: A. Japanese B. East_Asian C. Southeast_Asian D. South_Asian
 291 In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL_CHOICE:
 292 X’, replacing X with A, B, C, or D.
 293 **Model only.** {"name": "cultural_profile", "arguments": {"ingredient":
 294 "miso"}} **Model + Epicure.** FINAL_CHOICE: A
 295 **Epicure call.** cultural_profile({"ingredient": "miso"}). The answer key is
 296 FINAL_CHOICE: A.
 297 **Epicure result.** {"cuisines": {"East_Asian": {"range": {"p10": -0.1649, "p90":
 298 0.3039}, "score": 0.1625}, "Eastern_European": {"range": {"p10": -0.1466, "p90":
 299 0.1034}, "score": -0.1494}, "Japanese": {"range": {"p10": -0.1359, "p90": 0.1896},
 300 "score": 0.3227}, "Latin_American": {"range": {"p10": -0.1672, "p90": 0.1281},
 301 "score": -0.1538}, "Mediterranean": {"range": {"p10": -0.2323, "p90": 0.206}, "score":
 302 -0.1198}, "South_Asian": {"range": {"p10": -0.0808, "p90": 0.071}, "score": -0.0585},
 303 "Southeast_Asian": {"range": {"p10": -0.1625, "p90": 0.193}, "score": 0.0426},
 304 "Western_Atlantic": {"range": {"p10": -0.2304, "p90": 0.183}, "score": -0.1546}},
 305 "note": "Cosine similarity to each cuisine direction. Use p10/p90 to judge whether a score
 306 is typical or extreme.", "resolved": "miso"}
 307 These examples show exactly what the two measurements contain. Model only tests knowledge or
 308 inference about the hidden graph. Model + Epicure names the operation, so it tests whether the model
 309 can form valid arguments, read the returned structure, and preserve the answer contract. A correct
 310 choice with the wrong operation still earns answer credit, but the release separately reports whether
 311 the reference operation was matched. Across the run there are 606 projected EPICURE calls and 606
 312 model and task pairs matching the reference operation.

313 B Additional result figures

314 C Exact execution routes

315 D Protocol details

316 All arms request temperature 0, top- p 1, and seed 20260810 where the provider supports it. The final
 317 response budget is 2,048 tokens. Model + Epicure has one call round, one call total, a 4,096-byte
 318 result bound, and a 4,096-byte cumulative result bound. At most two provider attempts are allowed,
 319 and an ambiguously delivered request is never replayed. The final marker is parsed from plain text for
 320 transport portability; direct providers may use native structured generation internally and normalize
 321 to the same marker.

322 E Data fields

323 Each observation binds model ID, task ID, condition, source status, source and response hashes,
 324 returned model and provider, finish reason, response time, final answer, observed and expected choice,
 325 normal-parse flag, correctness, and a projected Epicure trace. Each trace event contains its name,
 326 structured arguments, error flag, response time, and result hash. The release does not require raw
 327 provider credentials or hidden service state.

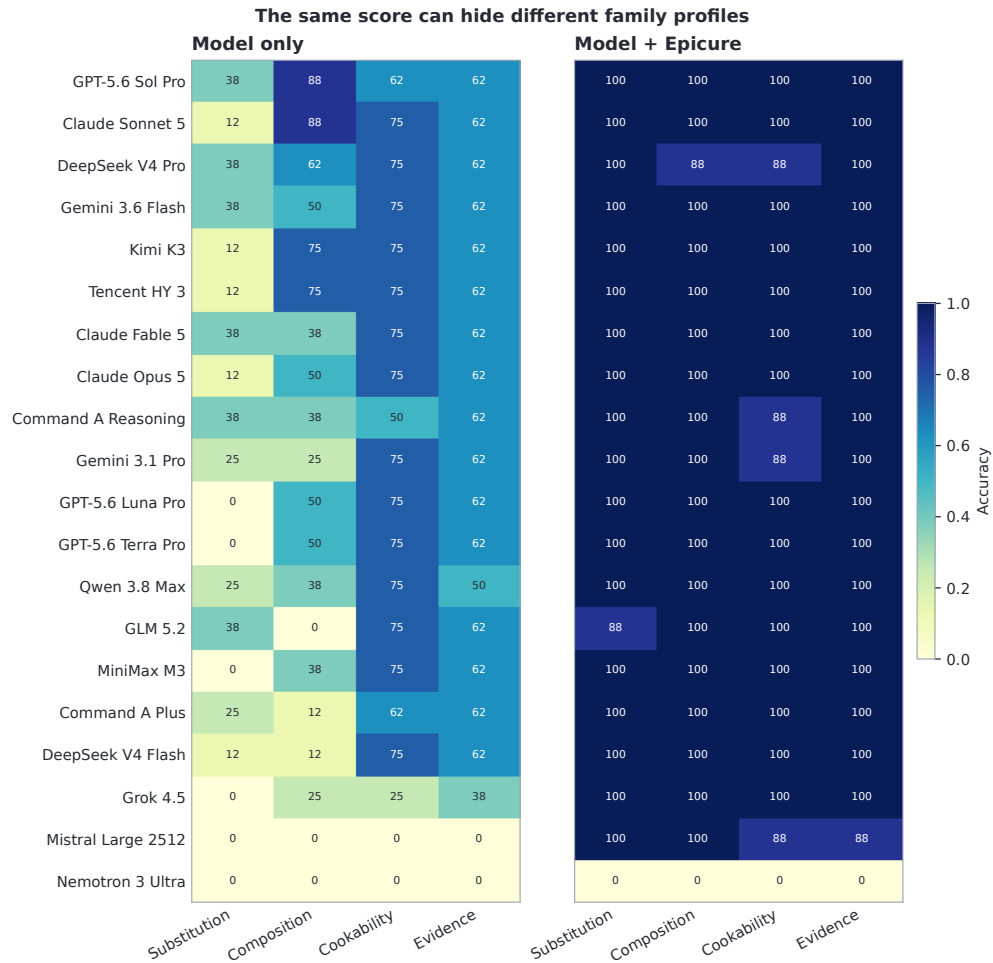


Figure 4: Model only and Model + Epicure accuracy by task family. Each value is a percentage over eight tasks. Similar FlavourBench Scores can come from different family profiles.

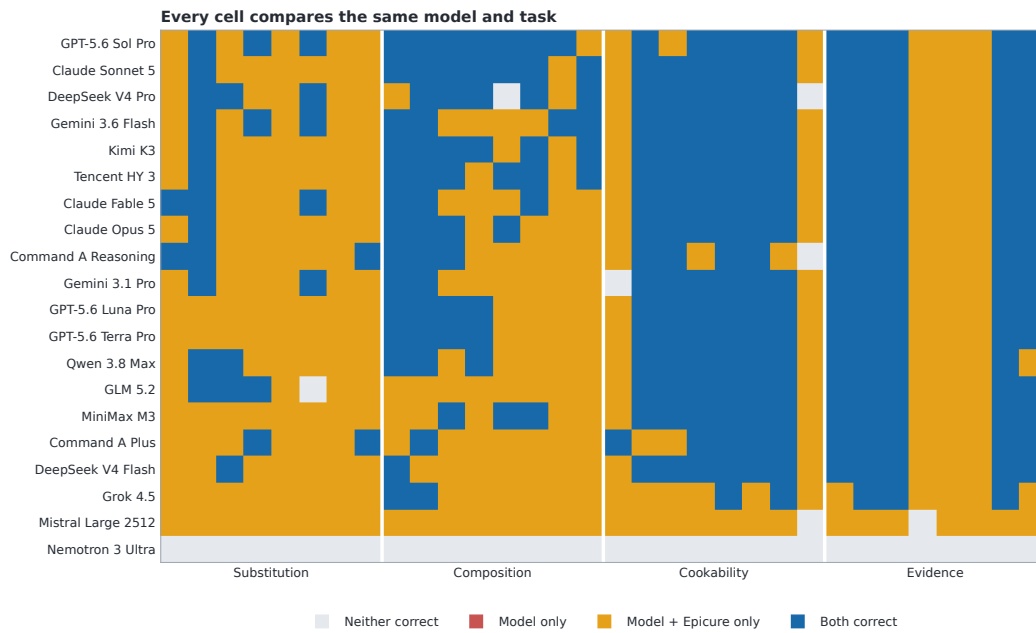


Figure 5: All 640 matched outcomes, ordered by FlavourBench Score and grouped by task family. The matrix shows which exact tasks produce Epicure Gain and where the assisted response regresses.

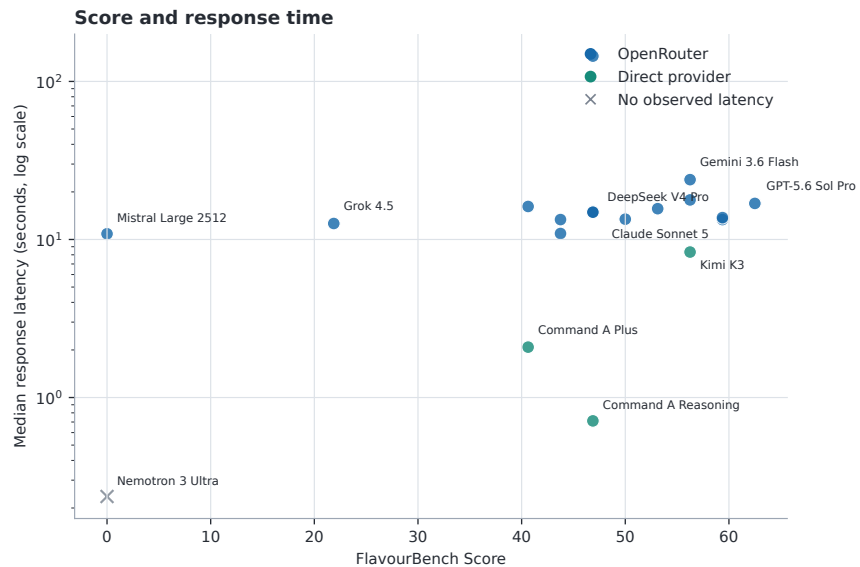


Figure 6: FlavourBench Score and response time. Response time uses a log scale; colour identifies direct-provider and OpenRouter execution. An × marks a route with no observed response time.

Slot	Model	Execution route
1	GPT-5.6 Sol Pro	openrouter: openai/flex
6	Claude Sonnet 5	openrouter: amazon-bedrock/claude-on-aws
13	DeepSeek V4 Pro	openrouter: baidu/fp8
8	Gemini 3.6 Flash	openrouter: google-vertex/global/flex
10	Kimi K3	kimi_direct: kimi-code-direct
18	Tencent HY 3	openrouter: baidu/fp8
4	Claude Fable 5	openrouter: amazon-bedrock
5	Claude Opus 5	openrouter: amazon-bedrock/claude-on-aws
3	GPT-5.6 Luna Pro	openrouter: openai/flex
2	GPT-5.6 Terra Pro	openrouter: openai/flex
11	Qwen 3.8 Max	openrouter: alibaba
20	Command A Reasoning	cohere_direct: cohere-direct
7	Gemini 3.1 Pro	openrouter: google-vertex/global/flex
15	MiniMax M3	openrouter: together
12	GLM 5.2	openrouter: coreweave/fp4
19	Command A Plus	cohere_direct: cohere-direct
14	DeepSeek V4 Flash	openrouter: cloudflare/fp8
9	Grok 4.5	openrouter: xai/zdr
17	Mistral Large 2512	openrouter: mistral
16	Nemotron 3 Ultra	openrouter: deepinfra/fp4

Table 6: All 20 model routes in the released leaderboard. Routes are selected before generation; automatic fallback is disabled.

References

- [1] Anonymous. Epicure: Navigating the Emergent Geometry of Food Ingredient Embeddings. *arXiv preprint arXiv:2605.22391*, 2026.
- [2] Anonymous. Epicure: Multidimensional Flavor Structure in Food Ingredient Embeddings. *arXiv preprint arXiv:2604.22776*, 2026.
- [3] Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael I. Jordan, Joseph E. Gonzalez, and Ion Stoica. Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 8359–8388. PMLR, 2024. URL <https://proceedings.mlr.press/v235/chiang24b.html>.
- [4] Donghee Choi, Mogan Gim, Donghyeon Park, Mujeen Sung, Hyunjae Kim, Jaewoo Kang, and Jihun Choi. CookingSense: A Culinary Knowledgebase with Multidisciplinary Assertions. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 3983–3996, Torino, Italia, 2024. ELRA and ICCL. URL <https://aclanthology.org/2024.lrec-main.354/>.
- [5] Cohere. Chat (V2) API Reference. <https://docs.cohere.com/v2/reference/chat>, 2026. URL <https://docs.cohere.com/v2/reference/chat>. Accessed 3 August 2026.
- [6] Cohere. Reasoning models. <https://docs.cohere.com/v2/docs/reasoning>, 2026. URL <https://docs.cohere.com/v2/docs/reasoning>. Accessed 3 August 2026.
- [7] Cohere. Tool use overview. <https://docs.cohere.com/v2/docs/tool-use-overview/>, 2026. URL <https://docs.cohere.com/v2/docs/tool-use-overview/>. Accessed 3 August 2026.
- [8] Tome Eftimov, Ana Gjorgjevikj, Matej Martinc, Gjorgjina Cenikj, Saso Dzeroski, and Barbara Koroušić Seljak. FoodBench-QA: Overview of the Shared Task on Grounded Food and Nutrition Question Answering. In *Proceedings of the Third Workshop on Patient-Oriented Language Processing (CL4Health) at LREC 2026*, 2026. URL <https://cs.ijs.si/publication/751>.
- [9] Neelansh Garg, Apuroop Sethupathy, Rudraksh Tuwani, et al. FlavorDB: A Database of Flavor Molecules. *Nucleic Acids Research*, 46(D1):D1210–D1216, 2018. doi: 10.1093/nar/gkx957.
- [10] Andong Hua, Mehak Preet Dhaliwal, Laya Pullala, Ryan Burke, and Yao Qin. NutriBench: A Dataset for Evaluating Large Language Models on Nutrition Estimation from Meal Descriptions. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://arxiv.org/abs/2407.12843>.
- [11] Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. LiveCodeBench: Holistic and Contamination Free Evaluation of Large Language Models for Code. *arXiv preprint arXiv:2403.07974*, 2024. URL <https://arxiv.org/abs/2403.07974>.
- [12] Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can Language Models Resolve Real-World GitHub Issues? In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=VTF8yNQm66>.
- [13] Song Jin, Juntian Zhang, Xun Zhang, Zeying Tian, Fei Jiang, Guojun Yin, Wei Lin, Yong Liu, and Rui Yan. DiningBench: A Hierarchical Multi-view Benchmark for Perception and Reasoning in the Dietary Domain. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 35289–35310. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.acl-long.1630. URL <https://aclanthology.org/2026.acl-long.1630/>.
- [14] Kimi Team. Kimi K3: Open Frontier Intelligence. *arXiv:2607.24653*, 2026. URL <https://arxiv.org/abs/2607.24653>.
- [15] Percy Liang, Rishi Bommasani, Tony Lee, et al. Holistic Evaluation of Language Models. *Transactions on Machine Learning Research*, 2023.
- [16] Bill Yuchen Lin, Yuntian Deng, Khyathi Chandu, Faeze Brahman, Abhilasha Ravichander, Valentina Pyatkin, Nouha Dziri, Ronan Le Bras, and Yejin Choi. WildBench: Benchmarking LLMs with Challenging Tasks from Real Users in the Wild. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/771155abaae744e08576f1f3b4b7ac0d-Abstract-Conference.html.

- [17] Mingyang Mao, Bhargav Rishi Mediseti, Utkarsh Grover, Tanvir Ibrahim, Wenyan Li, Tingting Zhang, and Xiaomin Lin. FAM-Bench: A Multimodal Benchmark for Condition-Aware Food-as-Medicine Reasoning. *arXiv preprint arXiv:2605.31410*, 2026. URL <https://arxiv.org/abs/2605.31410>.
- [18] Moonshot AI. Kimi code: Providers and models. <https://moonshotai.github.io/kimi-code/en/configuration/providers.html>, 2026. URL <https://moonshotai.github.io/kimi-code/en/configuration/providers.html>. Accessed 28 July 2026.
- [19] OpenRouter. Get Request and Usage Metadata for a Generation. <https://openrouter.ai/docs/api/reference/overview>, 2026. URL <https://openrouter.ai/docs/api/reference/overview>. Accessed 2026-07-15.
- [20] Shishir G. Patil, Huanzhi Mao, Fanjia Yan, Charlie Cheng-Jie Ji, Vishnu Suresh, Ion Stoica, and Joseph E. Gonzalez. The Berkeley Function Calling Leaderboard (BFCL): From Tool Use to Agentic Evaluation of Large Language Models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 48371–48392. PMLR, 2025. URL <https://proceedings.mlr.press/v267/patil25a.html>.
- [21] Yujia Qin, Shihao Liang, Yining Ye, et al. ToolLLM: Facilitating Large Language Models to Master 16000+ Real-world APIs. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=dHng200Jjr>.
- [22] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. GPQA: A Graduate-Level Google-Proof Q&A Benchmark. In *First Conference on Language Modeling*, 2024. URL <https://openreview.net/forum?id=Ti67584b98>.
- [23] Anka Reuel, Amelia Hardy, Chandler Smith, Max Lamparth, Malcolm Hardy, and Mykel J. Kochenderfer. BetterBench: Assessing AI Benchmarks, Uncovering Issues, and Establishing Best Practices. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/26889e8359e7ef8a7f5d77457364ca55-Abstract-Datasets_and_Benchmarks_Track.html.
- [24] Amaia Salvador, Nicholas Hynes, Yusuf Aytar, Javier Marin, Ferda Ofli, Ingmar Weber, and Antonio Torralba. Learning Cross-Modal Embeddings for Cooking Recipes and Food Images. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 3020–3028, 2017.
- [25] Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-3018. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/ad236edc564f3e3156e1b2feafb99a24-Abstract-Datasets_and_Benchmarks_Track.html.
- [26] Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Benjamin Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Jain, Khalid Saifullah, Sreemanti Dey, Shubh Agrawal, Sandeep Singh Sandha, Siddhartha Venkat Naidu, Chinmay Hegde, Yann LeCun, Tom Goldstein, Willie Neiswanger, and Micah Goldblum. LiveBench: A Challenging, Contamination-Limited LLM Benchmark. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [27] Zirui Wu, Xiao Liu, Jiayi Li, Lingpeng Kong, and Yansong Feng. Recipe2Plan: Evaluating Planning Abilities of LLMs for Efficient and Feasible Multitasking with Time Constraints Between Actions. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 4279–4301, Suzhou, China, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-emnlp.229. URL <https://aclanthology.org/2025.findings-emnlp.229/>.
- [28] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ -bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/1b126cc38b8638e07bef37e7b2bb72bf-Abstract-Conference.html.

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